

30th International Conference on Knowledge-Based and Intelligent Information & Engineering Systems (KES 2026)

Experimental Comparison of a Classical CNN and a Hybrid Quantum-Classical Model for Dermoscopic Skin Lesion Classification

Aida-Gabriela Pâncă, Maria Ciorăţanu, Ştefan-Răzvan Bălăucă, Adrian Iftene

“Alexandru Ioan Cuza” University of Iasi, Faculty of Computer Science

General Henri Mathias Berthelot Street, No. 16, 700259, Iasi, Romania

Abstract

Automated analysis of dermoscopic images can support benign-versus-malignant skin lesion classification, but the practical usefulness of hybrid quantum-classical models remains unclear under current hardware constraints. This paper presents an experimental comparison between a classical MobileNetV2-based baseline and a hybrid pipeline that combines convolutional feature extraction with a variational quantum circuit for binary classification on the publicly available HAM10000 dataset. In the hybrid approach, image features extracted by the classical backbone are projected to a low-dimensional latent space and encoded into a trainable quantum circuit. Both models are evaluated on disjoint train, validation, and test splits, with model selection for the quantum pipeline performed on the validation set. The classical baseline clearly outperformed the hybrid model. The selected hybrid configuration reached a balanced accuracy of 0.582 and an area under the receiver operating characteristic curve (AUC) of 0.610 on the held-out test set. We also validated inference on IBM Quantum backends, confirming compatibility with execution on real quantum hardware. Taken together, these results suggest that the hybrid pipeline is better viewed as a feasibility study under current noisy intermediate-scale quantum (NISQ) limitations rather than as a replacement for strong classical image classifiers.

© 2026 The Authors. Published by Elsevier B.V.

This is an open access article under the CC BY-NC-ND license (<http://creativecommons.org/licenses/by-nc-nd/4.0/>)

Peer-review under responsibility of the scientific committee of the KES International.

Keywords: dermoscopic image analysis; skin lesion classification; hybrid quantum-classical models; variational quantum circuits; HAM10000 dataset

1. Introduction

Automated analysis of dermoscopic images has become an important research direction for computer-aided skin lesion assessment, since early malignant cases may visually resemble benign lesions, while benign lesions may also exhibit atypical appearance [1, 2]. This high intra-class variability and inter-class similarity make binary lesion classification a difficult image analysis problem. Deep convolutional neural networks (CNNs) currently provide strong baselines for

E-mail address: aida.pnc9@gmail.com (Aida-Gabriela Pâncă).

such tasks because they can learn discriminative visual patterns directly from images and support effective transfer-based pipelines for downstream classification [1, 3, 4]. At the same time, there is growing interest in studying whether hybrid quantum-classical learning pipelines can offer useful experimental alternatives for classification under the constraints of noisy intermediate-scale quantum (NISQ) hardware [5–7]. In practice, fully quantum image processing remains difficult for realistic medical images because current hardware offers only a limited number of noisy qubits and limited connectivity. For this reason, many hybrid approaches assign high-dimensional feature extraction to a classical model and use the quantum component only as a lower-dimensional decision stage. This design makes it possible to study variational quantum circuits on non-trivial image data while staying within realistic resource constraints. However, the practical value of such systems depends not only on whether they can be trained, but also on whether they can be evaluated reproducibly, compared against strong classical baselines, and executed on real hardware rather than only on simulators.

In this paper, we study binary dermoscopic skin lesion classification on the publicly available HAM10000 dataset and compare two approaches under the same task definition [2]. The first is a classical MobileNetV2-based baseline. The second is a hybrid quantum-classical pipeline in which features extracted by the classical backbone are projected to a low-dimensional latent representation and then encoded into a parameterized quantum circuit that produces the final prediction score. In addition to held-out evaluation, we also validate inference on IBM Quantum backends in order to verify compatibility with execution on real quantum hardware. This work makes three main contributions. It presents a hybrid quantum-classical pipeline for benign versus malignant dermoscopic lesion classification that combines transfer-learned convolutional features, dimensionality reduction, and a variational quantum classifier. It provides a direct comparison against a strong classical baseline on HAM10000 under a clear train, validation, and test protocol. It also demonstrates inference on IBM Quantum backends and frames the resulting system as a feasibility study under current hardware limitations.

2. Related Work

Deep convolutional models are well-established baselines for dermoscopic lesion classification, and transfer-based pipelines remain a common choice in medical image analysis because they can benefit from strong visual representations learned from large-scale image data [1, 3]. On public benchmarks such as HAM10000, these models provide an important reference point for evaluating alternative architectures [2].

In quantum machine learning, practical image classification studies generally rely on hybrid rather than fully quantum pipelines. Parameterized quantum circuits have been studied as trainable models, while hybrid transfer-learning settings use classical feature extractors to reduce the dimensional burden before quantum processing [4, 6, 8]. This design is especially relevant in medical imaging, where current hardware limitations make direct quantum processing of high-resolution images impractical.

Recent hybrid quantum-classical studies for skin lesion analysis follow this general direction through different architectural choices, including MobileNetV2-based hybrid components, hybrid deep-learning and quantum workflows, quantum attention mechanisms, and comparative evaluations of quantum machine learning models on HAM10000 [9–12]. Related medical imaging work has also explored distributed hybrid quantum convolutional models under limited-qubit constraints [13]. Together, these studies show that hybrid quantum-classical models are being investigated as experimental alternatives for medical image classification, although their practical advantage over strong classical baselines remains an open question under current hardware constraints.

Compared with these studies, our work focuses on a controlled late-fusion setting in which MobileNetV2 is used as the classical feature extractor and the variational quantum circuit is evaluated as a compact decision layer operating on the reduced feature representation. The task definition, data split, and baseline comparison are kept explicit, which allows the behavior of the quantum decision module to be analyzed under the same HAM10000 evaluation protocol. Rather than proposing a more complex hybrid architecture, the contribution lies in the controlled comparison with a strong classical baseline, the explicit HAM10000 evaluation protocol, and the validation of the selected quantum decision module on IBM Quantum hardware.

3. Method

This section describes the task formulation, data processing pipeline, classical baseline, hybrid quantum-classical model, and inference setup used in our study.

3.1. Task and dataset

We study a binary dermoscopic image classification task in which each input image must be assigned to one of two classes, *benign* or *malignant*. The experiments are based on the HAM10000 dataset [2], a widely used public dermoscopic benchmark containing seven diagnostic categories. To obtain a binary setup aligned with our study, we aggregate the original labels into two super-classes. The malignant class contains melanoma (mel), basal cell carcinoma (bcc), and actinic keratoses (akiec), while the benign class contains melanocytic nevi (nv), benign keratosis-like lesions (bkl), dermatofibroma (df), and vascular lesions (vasc). Representative examples are shown in Fig. 1. All images are processed through the same preprocessing pipeline before classification. Inputs are resized to 224×224 pixels and pixel values are normalized to the $[0, 1]$ range. During training, we apply lightweight data augmentation through random rotations and flips in order to improve robustness to image variability. The same task definition and input representation are used for both the classical and hybrid models so that the comparison remains controlled and directly interpretable.

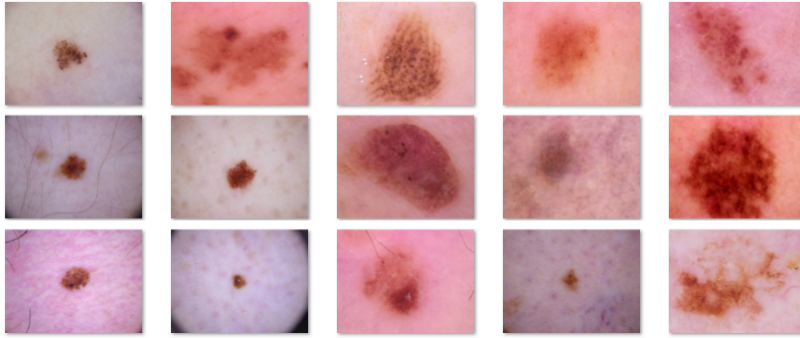


Fig. 1: Representative dermoscopic examples from HAM10000.

3.2. Classical baseline

As a classical reference model, we use a convolutional neural network based on MobileNetV2 [3]. The architecture was selected because it provides a strong trade-off between representational capacity and computational efficiency and is well suited to transfer learning on image classification tasks. We initialize the model with ImageNet pretrained weights and adapt it to the binary dermoscopic classification problem [4]. The convolutional backbone is used as a feature extractor, and the final classification head is replaced with a binary decision layer. Given an input image I , the model outputs a probability score $p_{\text{cnn}} \in [0, 1]$ for the malignant class, and the predicted label is obtained by thresholding this score. This model provides the main classical baseline for the comparison and also supplies the feature representation used by the hybrid pipeline.

3.3. Hybrid quantum-classical pipeline

The hybrid model follows a late-fusion design in which image understanding remains primarily classical, while the final decision stage is delegated to a variational quantum circuit (VQC). The pipeline begins with a frozen MobileNetV2 backbone that extracts a high-dimensional feature vector $f(I) \in \mathbb{R}^{1280}$ from the input image I . Because current quantum hardware cannot process such a representation directly, we reduce the feature dimension before quantum encoding. Concretely, the extracted features are standardized and projected with principal component analysis to a vector $z \in \mathbb{R}^n$, where n matches the number of qubits used by the quantum circuit. We then apply an element-wise hyperbolic tangent transformation, $x = \tanh(z)$, so that the resulting values lie in a bounded range suitable for angle encoding. More explicitly, if $f(I)$ denotes the feature vector extracted from the input image I , then the reduced latent representation is obtained as

$$z = \text{PCA}(\text{Scaler}(f(I))). \quad (1)$$

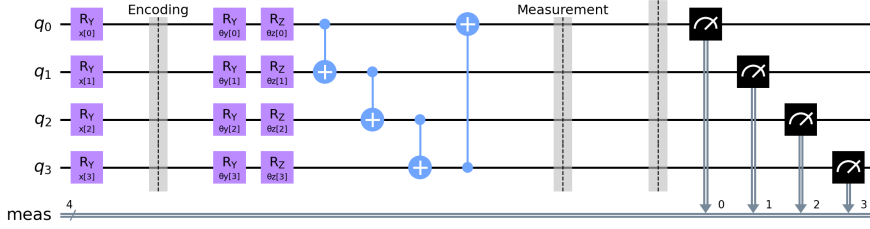


Fig. 2: Quantum circuit architecture: The data obtained after PCA is embedded into the initial quantum state via R_Y rotation gates. The trainable ansatz uses parametric R_Y and R_Z rotations and CNOT-based entanglement.

Here, $z \in \mathbb{R}^n$ and n is chosen to match the qubit budget of the quantum circuit. The encoded vector is then bounded through

$$x = \tanh(z). \quad (2)$$

This ensures that each component of x can be mapped to a stable rotation angle during quantum data encoding. The vector x is embedded into a parameterized quantum circuit implemented in Qiskit. Let $U(\mathbf{x}, \theta)$ denote the trainable circuit obtained after encoding the reduced feature vector $\mathbf{x}$ and applying the variational layers with parameters θ . The circuit acts on the initial state $|0\rangle^{\otimes n}$. For each qubit, the corresponding feature value is embedded through an angle-based rotation of the form

$$R_Y\left(\frac{\pi}{2} x_i\right). \quad (3)$$

Geometrically, this operation corresponds to a rotation of the corresponding qubit state on the Bloch sphere according to the bounded PCA component. Since $x_i \in [-1, 1]$, the encoding angle is restricted to $[-\pi/2, \pi/2]$. This bounded encoding keeps the input representation stable, but it also constrains the region of the quantum state space that can be reached directly from each feature. The representation is further limited by the qubit budget, since only n compressed components are encoded from the original MobileNetV2 feature vector. The trainable rotations and CNOT entanglement then determine how these encoded states are combined, and data re-uploading allows the same input features to influence multiple circuit layers. In this setting, the VQC acts as a low-dimensional nonlinear decision module, where the trained rotations and entanglement shape the decision boundary that is read out through Pauli-Z expectation values in the compressed quantum feature space.

After data encoding, the circuit applies trainable variational layers composed of single-qubit rotations and entangling gates. In our implementation, the trainable layers use R_Y and R_Z rotations together with CNOT-based entanglement as illustrated in Figure 2. The entanglement pattern can be configured as ring or linear, depending on the selected circuit setting. To increase expressiveness, the encoded input can be reintroduced across layers through data re-uploading [14]. This design follows the broader family of parameterized quantum models studied in quantum machine learning [6, 8].

3.4. Prediction rule

To obtain a scalar decision signal, we measure the mean Pauli-Z observable

$$O = \frac{1}{n} \sum_{i=1}^n Z_i. \quad (4)$$

The circuit output is then summarized through the expectation value

$$e = \langle 0 | U^\dagger(\mathbf{x}, \theta) O U(\mathbf{x}, \theta) | 0 \rangle. \quad (5)$$

Here, $e \in [-1, 1]$. We convert this quantity to a probability-like score for the malignant class as

$$p_{\text{vqc}} = \frac{e + 1}{2}. \quad (6)$$

The final predicted label is obtained through a calibrated threshold τ :

$$\hat{y} = \begin{cases} \text{malignant}, & p_{\text{vqc}} \geq \tau, \\ \text{benign}, & p_{\text{vqc}} < \tau. \end{cases} \quad (7)$$

This formulation defines the VQC score and the decision threshold used consistently during validation, test evaluation, and hardware inference.

3.5. Training and reproducible inference

Training of the hybrid model is performed offline. The circuit parameters are optimized with Simultaneous Perturbation Stochastic Approximation, which is well-suited to noisy optimization settings because it estimates gradients with only two objective evaluations per update step, independently of the number of trainable parameters. In our case, this makes the procedure compatible with shot-based variational optimization under hardware-inspired constraints. To support reproducibility, the trained pipeline is stored through a compact set of artifacts. These include the fitted feature transformation components used before quantum encoding, the learned circuit parameters, the selected decision threshold, and configuration metadata describing the retained setup. Concretely, the stored artifacts comprise the fitted scaler and PCA transformer, the optimized quantum parameters, the validation-selected threshold, and the associated configuration files. This artifact-based organization makes it possible to reload the same hybrid decision pipeline for repeated evaluation and for real-hardware inference without retraining.

3.6. Real-hardware inference on IBM Quantum

In addition to simulator-based experiments, we validate the hybrid inference pipeline on IBM Quantum hardware. To run the model on a real backend, the parameterized circuit must first be transpiled to operations supported by the selected device and mapped to a valid hardware layout. In practice, we use Qiskit's preset pass managers to generate circuits compatible with the selected backend instruction set and execute shot-based estimation through IBM Runtime primitives. The overall inference flow remains unchanged. The classical backbone extracts image features, the reduced representation is encoded into the variational circuit, and the final prediction is computed from shot-based measurement statistics. Execution on real hardware introduces practical constraints that are not fully captured by local simulation, including native gate restrictions, qubit connectivity limitations, sampling noise, and queue-dependent latency. For this reason, real-device runs are used as a feasibility and compatibility validation step rather than as the main environment for iterative model development. The purpose of this experiment is therefore not to claim a hardware-level performance advantage, but to show that the proposed hybrid decision module remains executable under present-day quantum hardware constraints. The same feature encoding and trained VQC parameters are preserved across simulator-based and hardware-based inference, with only the execution backend changing.

4. Experimental Setup

This section summarizes the evaluation protocol used to compare the classical baseline and the hybrid quantum-classical model. Our goal is to keep the comparison controlled by using the same binary task definition, the same dataset source, and clearly separated train, validation, and test splits.

4.1. Evaluation protocol

We consider a binary dermoscopic image classification task on HAM10000, with the benign versus malignant label grouping described in Section 3. The evaluation is performed using disjoint train, validation, and test splits. Model development and selection are carried out on the training and validation data, while final performance is reported only on the held-out test split. This protocol is especially important for the hybrid model, since its decision threshold is selected on the validation split and then applied unchanged at test time. To keep the comparison meaningful, both models operate on the same task and use consistent label definitions. At the same time, the comparison should be interpreted carefully. The hybrid model does not replace the full visual learning pipeline, because it reuses CNN-derived features and evaluates the quantum component as a compact decision layer on top of a reduced latent representation.

4.2. Metrics

We report standard classification metrics for the hybrid model on both validation and test data. In particular, we use accuracy, balanced accuracy, F1 score, and the area under the receiver operating characteristic curve (AUC). These metrics provide complementary views of model behavior. Accuracy summarizes overall correctness, balanced accuracy is useful when class balance must be treated more carefully, the F1 score reflects the trade-off between precision and recall, and AUC captures ranking quality across decision thresholds. In addition to scalar metrics, we also analyze confusion matrices on the held-out test set in order to examine the distribution of false positives and false negatives. This is particularly useful in the present setting, where the hybrid classifier operates under limited-dimensional quantum representations and stochastic optimization noise.

4.3. Hybrid model configuration

The hybrid classifier combines a frozen MobileNetV2 feature extractor with dimensionality reduction and a variational quantum circuit. Among the configurations explored during development, the selected setting uses $n_q = 12$ qubits, four variational repetitions, data re-uploading equal to 2, ring entanglement, and a total of $n_\theta = 192$ trainable parameters. The quantum model is evaluated through a shot-based estimator with 2048 shots. Circuit parameters are optimized with SPSA (Simultaneous Perturbation Stochastic Approximation) for 500 iterations and 5 random restarts. For this configuration, the best restart achieved its best validation score at iteration 225. Since the optimization is stochastic and the objective is evaluated through finite sampling, run-to-run variability is expected and is treated as part of the practical behavior of the method rather than as an implementation artifact.

4.4. Threshold selection and model selection

For the hybrid classifier, the raw circuit output is converted into a probability-like score, and the final binary decision depends on a threshold selected on the validation split. Threshold selection is therefore part of the model selection procedure and is completed before the test set is used. In the retained configuration, the chosen threshold is $\tau = 0.53$, selected on the validation set with balanced accuracy as the target criterion. This protocol ensures that the reported test performance corresponds to a fixed model and a fixed decision rule. It also keeps the evaluation aligned with the intended interpretation of the paper, namely a controlled feasibility study of a hybrid decision module under realistic NISQ-oriented constraints.

4.5. Classical baseline setting

As a reference system, we use a transfer learning baseline based on MobileNetV2. The model is trained for the same binary task and evaluated under the same split structure used for the hybrid model. This baseline serves as the main comparison point because convolutional neural networks remain the strongest practical family of models for dermoscopic image classification, and because the hybrid model itself relies on MobileNetV2-derived features before quantum processing.

4.6. Simulator-based evaluation and IBM Quantum validation

Most model development is conducted through simulator-based or local shot-based evaluation, which allows repeated experimentation during optimization and validation. After model selection, we additionally validate inference on IBM Quantum hardware by transpiling the retained circuit to backend-compatible operations and executing the same trained decision flow with shot-based measurements. This hardware experiment is used as a feasibility and compatibility check, not as the main environment for model development or as evidence of a hardware-level performance advantage.

5. Results and Discussion

Using the evaluation protocol described in the previous section, we compare the classical MobileNetV2 baseline with the hybrid quantum-classical pipeline on the binary benign versus malignant HAM10000 task. The main finding is that the classical baseline substantially outperforms the hybrid model in predictive performance, while the hybrid pipeline is better interpreted as a feasibility study under present noisy intermediate scale quantum constraints.

5.1. Hybrid model performance

The hybrid classifier is evaluated using the best retained configuration described in Section 4. Threshold selection is performed on the validation split. In the retained configuration, the selected threshold is $\tau = 0.53$, chosen to optimize balanced accuracy on validation data. With this threshold, the hybrid model reached accuracy = 0.64, balanced accuracy = 0.64, F1 score = 0.65, and AUC = 0.62 on the validation split, and accuracy = 0.582, balanced accuracy =

0.582, F1 score = 0.589, and AUC = 0.610 on the held-out test split. These values indicate that the hybrid model performs above chance level, but remains far from the predictive strength expected from a strong classical dermoscopic classifier. These results show that the full hybrid decision pipeline can be trained, selected on validation data, and evaluated consistently on held-out test images under realistic dimensionality and hardware-related constraints.

5.2. Comparison with the classical baseline

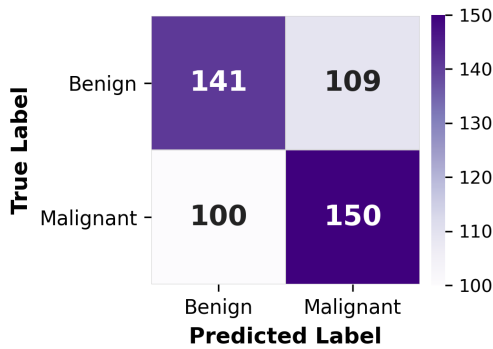
As a reference system, we evaluate a transfer learning baseline based on MobileNetV2 for the same binary task. The classical model achieved stronger performance, reaching 95.46% test accuracy. The observed gap is consistent with differences in representation and optimization. The classical baseline uses a high-capacity visual representation learned from large-scale image data and optimizes the binary classifier in a feature space adapted to visual discrimination. In the hybrid pipeline, the image is also represented first by MobileNetV2, but the resulting 1280-dimensional feature vector is compressed to a small number of PCA components before quantum encoding. This reduction is necessary for compatibility with the available qubit budget, while it can also remove fine-grained information that is useful for lesion discrimination. The quantum decision layer then operates on this compressed representation using finite-shot expectation estimates and SPSA optimization, which adds sampling noise and run-to-run variability to the decision process. The comparison therefore evaluates a compact VQC as a decision layer on top of classical features under the same HAM10000 protocol, rather than a fully quantum replacement for visual feature learning.

5.3. Error analysis

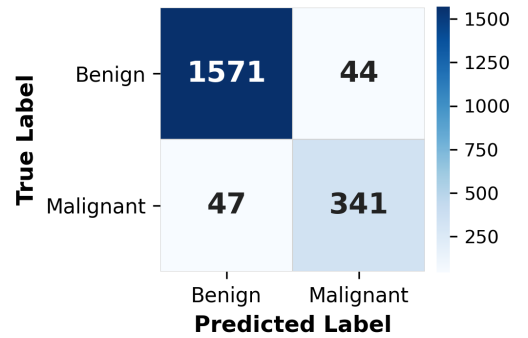
To better understand the behavior of the hybrid model, we analyze its confusion matrix on the held-out test set. The test confusion matrix for the retained hybrid configuration is

$$\begin{bmatrix} 141 & 109 \\ 100 & 150 \end{bmatrix},$$

where rows correspond to true labels and columns correspond to predicted labels. This distribution shows that the model produces a non-trivial number of both false positives and false negatives, which is consistent with the moderate scalar performance metrics reported above.



(a) Confusion matrix heatmap for the hybrid VQC model.



(b) Confusion matrix heatmap for the classical MobileNetV2 model.

Fig. 3: Confusion matrix comparison on the HAM10000 test split.

The heatmap in Figure 3a shows that the hybrid classifier produces errors in both classes, indicating that its decision boundary remains less stable and less discriminative than that of the classical baseline. By contrast, the classical baseline confusion matrix is more concentrated on the diagonal, reflecting stronger class separability in the CNN model (Figure 3b). This contrast is important for interpretation. The hybrid model demonstrates that a quantum decision layer can operate on compressed dermoscopic features and produce non-random predictions, but the resulting classifier still makes substantially more mistakes than the classical alternative.

5.4. Run-to-run variability and practical limitations

Across repeated VQC training runs, we observed noticeable variability. This behavior was influenced by finite-shot sampling noise, the stochastic nature of SPSA optimization, and architectural choices such as circuit depth, entanglement pattern, and data re-uploading. In our experiments, the best validation performance was in the range of 0.64–0.65 balanced accuracy, whereas other runs converged closer to chance level depending on initialization and hyperparameter choices. These results indicate that training stability is a practical limitation of the hybrid pipeline, in addition to the representational constraints imposed by the compressed quantum-compatible feature space. The repeated restarts give an initial indication of this variability. A broader ablation over qubit count, circuit depth, entanglement pattern, data re-uploading, and optimizer settings could further isolate the effect of each design choice, while additional independent seeds could support a more formal statistical robustness analysis. Accordingly, the hybrid results should be interpreted as a controlled feasibility study of a compact quantum-classical decision layer on dermoscopic data under current hardware constraints.

5.5. Validation on IBM Quantum hardware

Beyond simulator-based evaluation, we also validate inference on IBM Quantum backends. This step is important because it confirms that the retained hybrid decision pipeline is not restricted to offline simulation. For real-device execution, the parameterized circuit must be transpiled to hardware-supported operations and mapped to a valid backend layout before inference is run. The overall decision flow remains unchanged across simulator-based and hardware-based execution. In both cases, the classical backbone extracts image features, the reduced latent representation is encoded into the variational circuit, and the final score is computed from shot-based measurements. The hardware experiment therefore serves as a compatibility and feasibility validation step. It shows that the trained hybrid decision module remains executable on present-day quantum hardware, even though real-device noise, queue-dependent latency, and backend constraints make hardware execution unsuitable as the main environment for iterative model development. This result confirms that the retained hybrid model is operational beyond simulation and can be executed for inference on real IBM Quantum backends while preserving the same mathematical decision flow. It should be interpreted as evidence of feasibility and hardware compatibility rather than as evidence of a hardware-level performance advantage.

5.6. Overall discussion

Taken together, the results support a clear conclusion. The classical MobileNetV2 baseline remains the stronger practical solution for benign versus malignant dermoscopic image classification on HAM10000. The hybrid quantum-classical model is weaker in predictive performance, more unstable across runs, and more constrained by low-dimensional encoding and shot-based optimization. At the same time, the study demonstrates an end-to-end hybrid workflow in which classical visual features are compressed, processed by a variational quantum circuit, and evaluated both in simulation and on real IBM Quantum hardware. The hybrid pipeline thus provides an experimental benchmark for hybrid quantum-classical dermoscopic classification under current NISQ constraints.

A clinical implication of these results is that the selected hybrid model should not be interpreted as suitable for clinical decision support in its current form. Its test performance is too limited for medical use, especially in a task where false negatives and false positives can both have important consequences. The value of the hybrid model in this study is therefore methodological rather than clinical, since it shows how a compact quantum decision layer behaves when coupled with classical dermoscopic features under current hardware constraints.

For future clinically oriented work, improved predictive performance would also need to be accompanied by stronger interpretability. Explainability methods such as Gradient-weighted Class Activation Mapping (Grad-CAM), saliency maps, or feature-attribution analyses could help clarify which image regions and latent features influence the classical and hybrid predictions. Such analyses would be especially important before considering any medical decision-support setting.

6. System Prototype and Engineering Validation

6.1. Prototype architecture and user flow

In addition to the model-level evaluation, we implemented an end-to-end prototype that exposes both approaches through a web interface and a backend API. A Streamlit frontend communicates with a FastAPI backend, allowing the user to upload a dermoscopic image, trigger inference, and inspect the outputs of the classical CNN and the variational

quantum classifier within the same workflow. The overall user flow is straightforward. The user selects or uploads an image, the backend applies preprocessing and inference, and the interface returns a side-by-side comparison report with predicted labels and associated scores. This setup also supported repeated experimental validation, since the same stored artifacts could be reloaded for reproducible inference across runs.

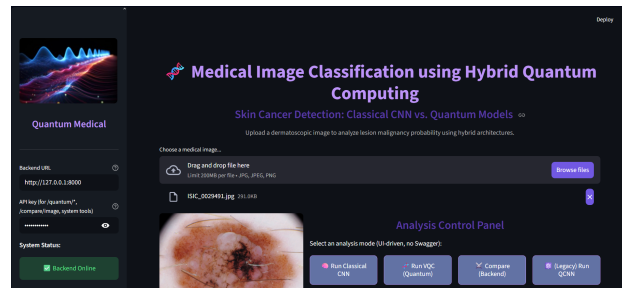


Fig. 4: Prototype interface for selecting a dermoscopic image before running inference.

6.2. Backend API and comparison workflow

The FastAPI backend exposes endpoints for single-model inference, comparative inference through `/compare/image`, and lightweight monitoring or debugging utilities. The comparison endpoint orchestrates both models asynchronously on the same uploaded image and returns a structured response containing per-model outputs, agreement indicators, score spread, and a selected best model according to score-based ranking. This design makes the comparison workflow explicit at the system level, rather than limiting it to offline metric reporting. It also supported inspection of cases in which the two models agreed or diverged on the same dermoscopic sample.



Fig. 5: Prototype interface showing the side-by-side comparison report for the classical CNN and the VQC model.

6.3. Operational support: caching, monitoring, and reproducibility

To improve responsiveness, the system uses prediction caching keyed by the uploaded image hash, with short-term persistence through a time-to-live policy. Repeated requests for the same image can therefore be served more efficiently, which was especially useful during testing and manual inspection of the compare workflow. The system also provided audit and monitoring support. In particular, inference requests could be tracked through an audit trail, while internal monitoring utilities supported inspection of API usage patterns and operational behavior. Together with the stored preprocessing and model artifacts described in Section 3, these mechanisms supported reproducible and operationally consistent evaluation.

6.4. Testing and non-functional evaluation

Beyond predictive performance, the prototype was validated through automated tests and non-functional checks. A total of 51 automated tests were executed, covering unit-level behavior, integration-level endpoint validation, and non-functional scenarios such as repeated requests and cache-aware execution paths. The recorded coverage report reached 73% overall coverage, with 90% coverage for the VQC-related module and 100% coverage for the classical CNN module. These values indicate that the most critical model-execution paths were directly exercised, even though some auxiliary utilities and interface-level code remained less covered. We also evaluated load behavior under repeated

requests. The resulting load-performance plot, shown in Figure 6, illustrates stable response times across increasing request loads and highlights the practical benefit of caching for repeated submissions.

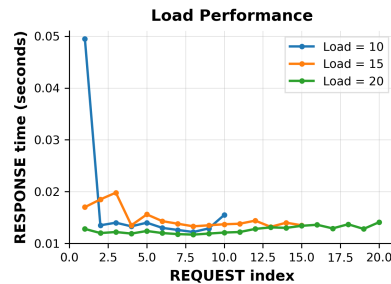


Fig. 6: Load-testing performance of the deployed API under repeated request loads.

7. Conclusion

This paper presented an experimental comparison between a classical MobileNetV2 baseline and a hybrid quantum-classical pipeline for binary dermoscopic skin lesion classification on HAM10000. The results show that the classical baseline remains clearly stronger in predictive performance, whereas the hybrid model achieves moderate above-chance performance under strong dimensionality and hardware-related constraints. The study also shows that a hybrid pipeline can be trained, selected using validation data, evaluated on a held-out test split, and executed for inference on IBM Quantum backends.

These findings should be interpreted in light of the constraints discussed in this study. The hybrid model relies on CNN-derived features, uses a qubit-limited compressed representation, and exhibits variability due to finite-shot sampling, SPSA optimization, and circuit-design sensitivity. Real-hardware inference serves as a feasibility validation on current IBM Quantum backends, while future work should address optimization stability, feature compression, circuit design, explainability, and noise-aware inference.

References

- [1] A. Esteva, B. Kuprel, R.A. Novoa, J. Ko, S.M. Swetter, H.M. Blau, and S. Thrun, “Dermatologist-level classification of skin cancer with deep neural networks,” *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.
- [2] P. Tschandl, C. Rosendahl, and H. Kittler, “The HAM10000 dataset, a large collection of multi-source dermoscopic images of common pigmented skin lesions,” *Scientific Data*, vol. 5, pp. 180161, 2018.
- [3] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.C. Chen, “MobileNetV2: Inverted Residuals and Linear Bottlenecks,” *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4510–4520, 2018.
- [4] A. Mari, T.R. Bromley, J. Izaac, M. Schuld, and N. Killoran, “Transfer learning in hybrid classical-quantum neural networks,” *Quantum*, vol. 4, pp. 340, 2020.
- [5] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, “Quantum Machine Learning,” *Nature*, vol. 549, no. 7671, pp. 195–202, 2017.
- [6] M. Cerezo, A. Arrasmith, R. Babbush, S.C. Benjamin, S. Endo, K. Fujii, J.R. McClean, K. Mitarai, X. Yuan, L. Cincio, and P.J. Coles, “Variational quantum algorithms,” *Nature Reviews Physics*, vol. 3, no. 9, pp. 625–644, 2021.
- [7] J. Preskill, “Quantum Computing in the NISQ era and beyond,” *Quantum*, vol. 2, pp. 79, 2018.
- [8] M. Benedetti, E. Lloyd, S. Sack, and M. Fiorentini, “Parameterized quantum circuits as machine learning models,” *Quantum Science and Technology*, vol. 4, no. 4, pp. 043001, 2019.
- [9] A.A. Hussein, A.M. Montaser, and H.A. Elsayed, “Skin cancer image classification using hybrid quantum deep learning model with BiLSTM and MobileNetV2,” *Quantum Machine Intelligence*, vol. 7, no. 2, pp. 66, 2025.
- [10] M. Frasca, I. Cutica, G. Pravettoni, and D. La Torre, “Optimizing melanoma diagnosis: A hybrid deep learning and quantum computing approach for enhanced lesion classification,” *Intelligence-Based Medicine*, vol. 12, pp. 100264, 2025.
- [11] P. Pandey, and S. Mandal, “A hybrid quantum-classical convolutional neural network with a quantum attention mechanism for skin cancer,” *Scientific Reports*, vol. 16, no. 1, pp. 1639, 2026.
- [12] S.S. Reka, H.L. Karthikeyan, A.J. Shakil, P. Venugopal, and M. Muniraj, “Exploring Quantum Machine Learning for Enhanced Skin Lesion Classification: A Comparative Study of Implementation Methods,” *IEEE Access*, vol. 12, pp. 104568–104584, 2024.
- [13] Y. Li, Z. Qia, Y. Lia, H. Yanga, R. Shanga, and L. Jiaoa, “A Distributed Hybrid Quantum Convolutional Neural Network for Medical Image Classification,” *arXiv preprint arXiv:2501.06225*, 2025.
- [14] A. Pérez-Salinas, A. Cervera-Lierta, E. Gil-Fuster, and J.I. Latorre, “Data re-uploading for a universal quantum classifier,” *Quantum*, vol. 4, pp. 226, 2020.